1. Title

"Resume Screening Machine Learning Model"

Mini - Project

Team Members :

1. Md Affan (INT2026-1905)
2. Md Danish (INT2026-2551)
3. Gunjan Damade (INT2026-2868)

2. Abstract

The rapid growth in job applications has made manual resume screening a time-consuming and error-prone process for recruiters. This project presents an **AI-based Resume Analyzer** that automates candidate role prediction and hiring decision support using Machine Learning techniques. The system analyzes candidate details such as skills, education, certifications, years of experience, salary expectations, and project count to predict the most suitable job role and determine whether the candidate should be shortlisted.

The system aims to reduce recruiter workload, improve hiring efficiency, and ensure objective decision-making. Experimental results demonstrate that the model achieves reliable predictive performance, making it suitable as a decision-support tool in recruitment processes.

3. Introduction

Recruitment is a critical process for any organization, as hiring the right candidate directly impacts productivity, innovation, and overall business success. With the increasing number of job applications received for every vacancy, manual resume screening has become time-consuming, inefficient, and prone to human bias. Recruiters often spend significant time shortlisting candidates, which can delay hiring decisions and increase operational costs.

To address these challenges, this project proposes an **AI-based Resume Analyzer** that leverages Machine Learning techniques to automate resume evaluation and support hiring decisions. The system analyzes candidate information such as skills, educational qualifications, certifications, years of experience, salary expectations, and project involvement. Based on these features, the system predicts the most suitable job role and provides a hiring recommendation (Hire/Reject).

This AI-powered solution aims to improve recruitment efficiency, reduce manual effort, and enhance objectivity in the screening process. By utilizing data-driven decision-making, the system supports recruiters in identifying suitable candidates more quickly and accurately.

4. Problem Statement

Organizations receive a large number of job applications for every open position. Manual resume screening is time-consuming, inconsistent, and prone to human bias. Recruiters may overlook qualified candidates or spend excessive time evaluating unsuitable profiles. Therefore, there is a need for an automated, intelligent system that can efficiently analyze resumes, predict suitable job roles, and assist in hiring decisions using data-driven techniques.

5. Objective

The main objectives of this project are:

1. To develop a machine learning-based system for resume analysis.
2. To predict the most suitable job role based on candidate features.
3. To provide a hiring recommendation (Hire/Reject).
4. To evaluate and compare multiple classification models.

6. Literature Review

Recent advancements in Artificial Intelligence and Machine Learning have significantly improved automated decision-making systems. Resume screening systems use Natural Language Processing (NLP) and classification algorithms to extract candidate features and predict suitability.

Studies show that algorithms such as Logistic Regression, Support Vector Machines (SVM), Random Forest, and Gradient Boosting perform well in classification-based recruitment systems. Evaluation metrics such as Precision, Recall, and F1-score are commonly used to measure performance, especially in imbalanced datasets.

Automated recruitment systems improve efficiency, reduce bias, and enhance accuracy compared to traditional manual screening methods.

7. Data set Description

Data Columns :

- Skills
- Education
- Certifications
- Experience (Years)
- Salary Expectations
- Projects Count
- Hiring Decision / Job Role

8. Data preprocessing

Data preprocessing is an essential step in the machine learning pipeline. Raw datasets often contain missing values, inconsistent formats, and categorical variables that cannot be directly used by machine learning algorithms. Therefore, preprocessing techniques are applied to clean, transform, and prepare the dataset's before training the models.

1. Data Cleaning

The first step involved cleaning the dataset to ensure data quality and consistency.

The following actions were performed:

- Removal of duplicate records .
- Handling missing values in features such as skills, certifications, or education.
- Ensuring numerical fields like experience, salary expectations, and project count contain valid numeric values.

2. Handling Missing Values

Some candidate profiles may contain incomplete information. Missing values were handled using appropriate techniques:

- Text fields such as **skills and certifications** were filled with default values if empty.
- Numerical features such as **experience or project count** were replaced with median or mean values where necessary.

3. Feature Selection

Relevant features were selected from the dataset to improve model performance. The following features were used:

- Skills
- Education
- Certifications
- Experience (Years)
- Salary Expectations
- Projects Count

Irrelevant or redundant attributes were removed to reduce model complexity and improve prediction accuracy.

4. Encoding Categorical Variables

Machine learning algorithms require numerical input. Therefore, categorical variables such as education, skills, and certifications were transformed into numerical representations.

Two common encoding techniques were used:

Label Encoding

Categorical values were converted into numerical labels.

Example:

Education	Encoded Value
B.Tech	0
M.Tech	1
MBA	2
PhD	3

One-Hot Encoding

Some categorical features were converted into binary columns to avoid unintended ordinal relationships.

Example:

Education_BTech	Education_MTech	Education_MBA
1	0	0

5. Feature Transformation

Text-based inputs such as **skills and certifications** were standardized by converting all characters to lowercase and removing unnecessary spaces. This ensures consistent representation of similar features.

6. Data Splitting

After preprocessing, the dataset was divided into training and testing sets using an **80:20 ratio**.

- **Training Data (80%)** – Used to train the machine learning models.
- **Testing Data (20%)** – Used to evaluate model performance on unseen data.

9. Exploratory Data Analysis

- Explored the dataset to understand its **structure, size, and data types**.
- Checked for **missing values and duplicate records**.
- Analyzed the **distribution of features** such as education, experience, skills, certifications, projects count, and salary expectations.
- Used visualization libraries like Pandas, Matplotlib, and Seaborn to generate charts and graphs.
- Studied the **relationship between features and hiring decision**.
- Identified **important factors influencing candidate selection**, such as experience, relevant skills, and certifications.

  The insights from EDA helped in **feature selection and improving model performance**.

10. Model Development

Model development involves building machine learning models to predict hiring decisions and recommend suitable job roles based on candidate information such as skills, education, experience, certifications, and projects.

10.1 Baseline Model

A baseline model was developed to establish an initial performance benchmark for the system. Simpler machine learning algorithms were used because they are easy to implement, interpret, and provide a reference point for evaluating more complex models.

Algorithms such as **Logistic Regression** and **Decision Tree** were used as baseline models. These models were trained using the preprocessed dataset containing candidate features.

The baseline model helped in:

- Understanding the relationship between input features and hiring outcomes.
- Establishing a **reference accuracy and F1-score**.
- Identifying limitations that could be improved using advanced models.

Although the baseline model provided reasonable predictions, its performance was limited when handling complex relationships within the dataset.

10.2 Advanced Model

To improve prediction accuracy and overall performance, more advanced machine learning models were implemented.

Algorithms such as **Random Forest** and **Gradient Boosting** were used. These models combine multiple decision trees to improve prediction accuracy and reduce overfitting.

The advanced model offered several advantages:

- Better handling of **non-linear relationships** in the data.
- Improved **prediction accuracy and F1-score**.
- More robust performance on unseen test data.
- Ability to capture complex patterns in candidate profiles.

After training, the models were evaluated using metrics such as **accuracy, precision, recall, and F1-score**, and the best-performing model was selected for deployment in the AI Resume Analyzer system.

11. Model Validation and Selection

Model validation and selection is an important step in the machine learning process. It ensures that the developed model performs well on unseen data and generalizes effectively. In this project, different machine learning models were evaluated and compared to select the best-performing model for the AI Resume Analyzer system.

Model Validation

To validate the performance of the models, the dataset was divided into **training and testing sets** using an 80:20 ratio. The training set was used to train the models, while the testing set was used to evaluate their performance on unseen data.

Various evaluation metrics were used to measure model performance, including:

- **Accuracy** – Measures the overall correctness of the model.
- **Precision** – Indicates how many predicted positive results were actually correct.
- **Recall** – Measures the model's ability to correctly identify positive cases.
- **F1 Score** – A balanced metric that combines precision and recall.

These metrics helped in understanding how well each model performs in predicting hiring decisions.

Model Selection

Multiple algorithms such as **Logistic Regression**, **Decision Tree**, and **Random Forest** were trained and evaluated. The performance of each model was compared using the evaluation metrics.

Among all the models, **Random Forest** achieved the highest performance in terms of accuracy and F1 score. It was able to handle complex relationships between candidate features such as skills, education, experience, and certifications.

Therefore, the **Random Forest model** was selected as the final model for deployment in the AI Resume Analyzer system due to its higher accuracy, robustness, and better generalization capability.

12. Deployment

Deployment is the final stage of the machine learning pipeline where the trained model is integrated into a working application so that users can interact with it and obtain predictions in real time.

The deployment workflow works as follows:

- The user enters candidate information in the web interface.
- The data is sent to the backend server using **Flask**.
- The server processes and formats the input data.
- The trained machine learning models generate predictions such as hiring decision and suitable job role.
- The results are displayed back to the user on the webpage.

The trained models were stored as serialized files using **Joblib or **Pickle**, which allows them to be loaded quickly during runtime.

13. Results and Business Impact

Results

The developed AI Resume Analyzer system was evaluated using various performance metrics to measure its effectiveness in predicting hiring decisions and recommending job roles. The models were tested on unseen data to ensure reliable performance.

The evaluation showed that advanced models such as **Random Forest** performed better compared to simpler models like **Logistic Regression** and **Decision Tree**. The selected model achieved higher accuracy and F1-score, indicating better balance between precision and recall.

Business Impact

The AI Resume Analyzer provides several benefits for organizations and recruitment teams:

- **Faster Recruitment Process:** Automates the initial screening of resumes, reducing manual effort.
- **Improved Decision Making:** Provides data-driven insights for selecting suitable candidates.
- **Time Efficiency:** HR teams can evaluate multiple candidates quickly.
- **Reduced Bias:** Machine learning models analyze candidates based on skills and qualifications rather than subjective judgment.
- **Better Talent Matching:** Recommends job roles that match the candidate's profile and experience.

14. Visuals and Screenshots

```python
from sklearn.metrics import classification_report, confusion_matrix

y_pred = model.predict(X_test)

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Confusion Matrix:
 [[ 25   0]
 [  7 114]]

Classification Report:
               precision    recall  f1-score   support

           0       0.78      1.00      0.88        25
           1       1.00      0.94      0.97       121

    accuracy                           0.95       146
   macro avg       0.89      0.97      0.92       146
weighted avg       0.96      0.95      0.95       146
```

1.1 Binary Classification Model Evaluation: Confusion Matrix and Classification Report”

```python
from sklearn.model_selection import GridSearchCV

param_grid = {
    "classifier__C": [0.1, 1, 10, 50],
    "classifier__kernel": ["linear"]
}

grid = GridSearchCV(
    model,
    param_grid,
    cv=5,
    scoring="f1",
    n_jobs=-1
)

grid.fit(X, y)

print("Best Parameters:", grid.best_params_)
print("Best F1 Score:", grid.best_score_)
```

```
Best Parameters: {'classifier__C': 50, 'classifier__kernel': 'linear'}
Best F1 Score: 0.989992735683485
```

1.2 “SVM Hyperparameter Tuning using GridSearchCV (Best Parameters & F1 Score)”

```
prediction = final_model.predict(sample_candidate)
probability = final_model.predict_proba(sample_candidate)

print("Prediction:", "Hire" if prediction[0] == 1 else "Reject")
print("Confidence:", max(probability[0]))
```

```
Prediction: Hire
Confidence: 0.9999999999997226
```

1.3 “Final Candidate Hiring Prediction with Model Confidence Score

15. Challenge & Limitation

During the development and implementation of the AI Resume Analyzer system, several challenges and limitations were encountered.

Challenges

- **Data Quality Issues:** The dataset contained inconsistent formats and missing values, which required additional preprocessing before training the models.

- **Feature Representation:** Converting textual information such as skills and certifications into numerical format suitable for machine learning models was challenging.

- **Model Selection:** Different algorithms had varying performance levels, making it necessary to test and compare multiple models before selecting the best one.

- **Data Imbalance:** Some job roles or hiring outcomes had fewer samples, which could affect the model's prediction accuracy.

- **Integration with Web Application:** Connecting the trained models with the web interface using **Flask** required careful handling of input data formats.

Limitations

- **Limited Dataset Size:** The accuracy of the model depends heavily on the amount and quality of training data. A small dataset may limit the model's ability to generalize.

- **Static Skill Analysis:** The system analyzes only predefined skills and may not fully capture emerging technologies or niche skill sets.

- **No Resume Parsing:** The system requires manual input of candidate details instead of automatically extracting information from resumes.

- **Model Bias:** If the training dataset contains bias, the predictions may also reflect those biases.

- **Scalability:** The current prototype is suitable for small-scale applications and may require optimization for large-scale recruitment systems.

16. Future Work

The AI Resume Analyzer system can be further improved by incorporating additional features and advanced technologies. Future enhancements can make the system more accurate, scalable, and useful for real-world recruitment processes.

- **Automatic Resume Parsing:** Integrate Natural Language Processing techniques to automatically extract candidate information directly from resumes instead of manual data entry.

- **Advanced Machine Learning Models:** Implement more advanced algorithms such as **XGBoost** or **Deep Neural Networks** to improve prediction accuracy.

- **Larger and Diverse Dataset:** Training the model on a larger and more diverse dataset will improve its ability to generalize and make more reliable predictions.

- **Real-time Recruitment Dashboard:** Develop an interactive dashboard for HR teams to visualize candidate insights and hiring trends.

- **Skill Gap Analysis:** The system can be extended to recommend skills that candidates should learn to improve their chances of getting hired.

- **Integration with Job Portals:** The system could be integrated with online recruitment platforms to automatically analyze candidate profiles.

- **Cloud Deployment:** Deploy the application on cloud platforms such as **Amazon Web Services** or **Google Cloud** for better scalability and accessibility.

17. Conclusion

The AI Resume Analyzer system was developed to assist in the recruitment process by using machine learning techniques to analyze candidate profiles and predict hiring decisions. The system evaluates important candidate attributes such as education, skills, certifications, experience, project count, and salary expectations to determine suitable job roles and hiring outcomes.

Different machine learning models were implemented and compared, including **Logistic Regression**, **Decision Tree**, and **Random Forest**. Among these, Random Forest achieved better performance in terms of accuracy and F1-score, making it the most suitable model for the application.

The trained model was successfully integrated into a web application using **Flask**, allowing users to input candidate details and receive predictions through an interactive interface.

Overall, the system demonstrates how machine learning can improve the efficiency of resume screening and support HR teams in making data-driven hiring decisions. With further improvements such as automatic resume parsing and larger datasets, the system can become a powerful tool for modern recruitment processes.